

SENTINEL

Statewide CCTV Integration Platform & ANPR Surveillance Engine

- **Event:** Gujarat CCTV Integration Hackathon 2026
- **Category:** Category 1 (Academic / Research / Startup)
- **Team Name:** KodeMatrix
- **Track Scope:** Model 1 (Central CCTV Registry & GIS Mapping) + Model 2 (Unified Viewing Platform & ANPR Alerting)

A production-grade platform running against the live Sentinel sandbox grid: 30 cameras onboarded and mapped, a continuous ANPR pipeline validated at 100% on legible footage, real-time watchlist alerting, and timestamped route reconstruction across cameras.

Platform Running - Live Dashboard:

 Sentinel Platform - Live Dashboard showing real CCTV feed, plate detections, and alerts

Slide 2a: The Problem

The Operational Gap in Statewide Surveillance

- Cameras are owned by multiple departments in diverse formats with no central registry.
 - Investigators asking "where has this vehicle been?" have no unified system to query across jurisdictions.
 - Watchlist matching, where present, relies on manual human monitoring across isolated displays.
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Slide 2b: What Sentinel Delivers

One platform. Every camera. Automatic.

- **Central Onboarding** - any catalogue camera, location provenance recorded
 - **Unified Viewing** - live multi-camera wall, no per-department software
 - **Continuous ANPR** - every plate, every camera, every timestamp
 - **Watchlist Matching** - exact-then-fuzzy trigram on every detection
 - **Real-Time Alerts** - WebSocket dispatch, severity levels, instant
 - **Route Reconstruction** - interactive map timeline with speed estimation
 - **Complete Auditability** - purpose-bound audit on every search and export
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Slide 3: System Architecture

 Sentinel System Architecture - ANPR Worker, FastAPI Backend, React UI, PostgreSQL

Technology Stack: Python 3.12+ / FastAPI | PostgreSQL 17 + PostGIS + pg_trgm | React + Vite + TypeScript | Leaflet | hls.js | ONNX Runtime (CPU)

Slide 4: Model 1 - Registry & GIS Mapping

Handling Real-World Data Gaps with Integrity

The sandbox catalogue publishes **only** `id` and `name` : no coordinates, no department, and no codec hints.

Our Solution

- Resolved all 30 camera locations from real site names published in the catalogue.
- Implemented strict location provenance precedence: hand-verified → Nominatim geocoding (rate-limited, cached, bounded to Gujarat) → district centroid → **null**.
- Stored `geo_source` and `geo_confidence` on every camera record so provenance travels into reports and UI displays.

What We Refused To Do

An early version placed Junagadh cameras in Surat. We caught it, and removed it.

GIS Capabilities: PostGIS geography points, GeoJSON export, department / status / confidence map layers, clustered markers, and click-through to live video tiles.

Slide 5: Model 2 - ANPR & Accuracy Engine

Local CPU Inference | Zero Third-Party API Dependence

Pretrained open-source models (YOLOv9-t detector, cct-s-v2 OCR) executing on **local CPU**. No video frame ever leaves the deployment; no external service sits in the alerting path.

Accuracy Achieved Through Domain-Specific Engineering

- 1. Track-Level Voting:** Vehicles remain visible across 20-60 frames. Every read votes per character weighted by OCR confidence, right-aligned on the 4-digit serial. Tracks reach 20-35 reads and accumulate 1.000 confidence.
- 2. Indian Plate Grammar:** Position rules make common OCR confusions deterministically correctable.

Real Example: OCR returned GJ01AB1234 . Position 2 must be a digit - O becomes 0 unambiguously. 7/7 pass rate on correction test suites. The same logic corrects five other common OCR confusions.

- 3. Overlay Rejection:** Detector filters out burnt-in timestamps, PTZ text overlays, and billboard signage before indexing.

Ground-Truth Benchmark: 6/6 plates recovered exactly, 0 false positives.

Live Sandbox Feed Findings: cam12 (Adalaj Toll Naka) is a toll plaza feed: the pipeline detected a real truck plate **25 times in 100 seconds**, recovering 8 of 10 characters at 5 px/character. Seven preprocessing variants confirmed that upscaling cannot restore detail missing from optics.

The constraint is optics, not software. Full detail in HLD §6.5.

Slide 6: Continuous ANPR Engine & Scalability

Inference Costs & Siting Realities

At 1920x1080 resolution, wide-area night feeds render number plates at 5-15 px wide. Full-frame inference at 384 px proposes zero candidates. Tiled inference on overlapping upscaled regions recovers plate regions at 15x CPU cost.

Benchmark on cam05 (20-core CPU, No GPU)	Full Frame	Tiled 2x3 @2x
Mean Inference Time	12.3 ms/frame	185.9 ms/frame
Est. Concurrent Streams per Machine	~251	~26
Plate Candidates (90s window)	0	8
Ground-Truth Plates Recovered	3/6	6/6

Analytics Policy Tiers

ANPR is not a uniform per-camera cost. It is a **policy choice regarding which cameras require analytics**:

Analytics Tier	Share of Estate	Deployment Workload
Tier 1: Continuous ANPR	5-10%	Highways, border posts, ANPR-sited toll plazas
Tier 2: Event-Triggered	20-30%	Urban junctions (motion / signal triggered)
Tier 3: Registry & View	60-75%	Full video coverage without analytics compute cost

Slide 7: Correlation, Alerting & Route Reconstruction

Watchlist Matching: Exact + Fuzzy Trigram Search

Every detection is matched against active watchlists before database transaction commit: exact match first, followed by `pg_trgm` trigram similarity > 0.7. Matches generate instant WebSocket alerts with severity metadata.

Why Fuzzy Search is Critical: ANPR OCR will occasionally misread a character. Exact-only matching misses genuine detections. Verified: searching `GJ01AB1234` correctly retrieves records stored as `GJØ1AB1234`.

Route Reconstruction

Sightings ordered chronologically - great-circle distance & elapsed time calculated - implied speed estimated - **physically impossible transitions (>150 km/h) flagged** and surfaced to operators rather than hidden.

Flagged transitions notify investigators of potential plate misreads or cloned registration plates, providing actionable intelligence.

Slide 8: Security, Privacy & Accountability

JWT with three roles, enforced as a dependency on each route - verified: a viewer token returns 403 on the audit trail and 200 on the camera registry.

- **Access Control:** state_admin / dept_operator / viewer roles on every protected endpoint
- **Audit Trail:** actor from verified token, purpose and case reference on every search and export
- **Data Minimisation:** plate text, timestamp, camera, crop only - no continuous video retained
- **Data Residency:** 100% local inference, zero frames leave the deployment
- **DPDP Act 2023:** purpose-bound access, role restrictions, complete access trail

Closed Government Systems (VAHAN, SARTHI, eGujCop)

Contract-first adapter pattern: request/response models defined, mock adapters with realistic records, exact activation steps documented for when production credentials arrive.

Slide 9: Scaling to 80,000 Cameras

Measured Bottleneck (not projected)

Concurrent Streams	Succeeded	Wall Time	Our Server CPU (20 Cores)
2	2 / 2	12 s	4%
4	4 / 4	26 s	4%
6	6 / 6	41 s	5%
8	7 / 8	88 s	4%

Key Insight: Our server sat at 4% CPU while the gateway took 73s to accept 8 connections. The bottleneck is centralized ingress - not compute. This is the 80,000-camera problem at a scale of eight.

Statewide Edge-First Architecture

 Statewide Edge-First Architecture - District Edge to Regional DC to State Core

Restricting WAN backhaul to metadata, alerts, and requested clips reduces wide-area bandwidth demand from **360 Gbps to 8.7 Mbps** - two to three orders of magnitude.

Slide 10: Platform Capabilities & Transparent Evaluation

Fully Demonstrated Capabilities

- 30 sandbox cameras onboarded with full location provenance
- Interactive GIS map with department, status, and confidence filters
- Unified video wall featuring HLS grid and WebRTC hero tiles
- Continuous ANPR pipeline with tiled inference, track voting, and grammar correction
- Exact, partial, and fuzzy plate search
- Automatic watchlist matching with real-time WebSocket alerts
- Route reconstruction with speed calculation and impossible transition flagging
- Output report generation in XLSX and PDF formats
- Purpose-bound audit trail logging
- 57 automated tests, 100% accuracy on legible ground-truth feeds

Transparent Limitations

Full transparent limitations - sandbox camera yield, geocoded location accuracy, single-node status - are documented in HLD Section 12.

Slide 11: Impact on Policing & Strategic Roadmap

Operational Transformation

Today, an investigator manually contacts multiple departments, scrubs hours of footage, and assembles timelines over several days. With Sentinel, they type a vehicle registration once - every sighting across all connected cameras appears on an interactive map within seconds.

Strategic Roadmap

1. **Phase 1:** Pilot deployment across 1 district (50-200 cameras) at toll plazas and checkpoints.
 2. **Phase 2:** Multi-VMS adapter rollout across urban junctions (Model 3 integration).
 3. **Phase 3:** Live VAHAN API credential activation and edge node expansion.
 4. **Phase 4:** Statewide edge node deployment across 80,000 cameras (Model 4 scale-out).
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Slide 12: Verification & Reproducibility

Claim / Benchmark	Verification Command / URL
ANPR Throughput Benchmark	<code>python anpr_worker.py --benchmark</code>
Ground-Truth 6/6 Accuracy Test	<code>python tools/make_sample_feed.py --validate</code>
Grammar & System Test Suite	<code>python -m pytest tests/ -q</code> (57 tests passing)
Ingest & Provenance Endpoint	<code>GET /api/v1/ingest/status</code>
Audit Trail API	<code>GET /api/v1/analytics/audit</code>
Interactive OpenAPI Docs	<code>http://localhost:8000/api/docs</code>